MILITARY SPECIFICATION

BORING AND TURNING MACHINES, VERTICAL, COMPUTER NUMERICAL CONTROL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers computer numerically controlled vertical boring and turning machines with equipment and accessories specified herein.

1.2 Classification. The boring and turning machines shall be of the following types. The machine types to be furnished shall be as specified (see 6.2.1).

Type I - Single rail head, turret type
Type II - Single rail head, ram type
Type III - Single turret head and side head
Type IV - Single ram head and side head
Type V - Turret head, ram head, and side head
Type VI - Two rail heads, ram type
Type VII - Two ram heads and side head

Sizes *: 36- through 118-inch table diameters

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Emission and Susceptibility, Test Methods For.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified,

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the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B5.51M - Preferred SI Units for Machine Tools.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA) STANDARDS

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.

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RS-244 - Character Code for Numerical Machine Tool Control.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools.

RS-274 - Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.

RS-281 - Construction Standards - Numerical Machine Tool Control

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

RS-494 - 32-Bit Binary CL Exchange (BCL) Input Format for Numerically Controlled Machines.

(Applications for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI)

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in accordance with ANSI B5.51.M shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2. Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Provisions shall be made to prevent damage to the feedback device in the event of overtravel.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Controls. All controls required for normal operation shall be located convenient to the operator at his normal work station.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and

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assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR 1910 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.5 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.7 Environmental protection. The machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of

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these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein, except for leadscrews which may be the manufacturer's standard.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Gears. All gears and pinions of spindle and axis drive train shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a base, work-holding table, vertical column or columns, crossrail, crossrail head units, side head, if required, and machine control unit. The machine axis and motion nomenclature shall conform to the requirements of EIA-RS-267. The machine shall be equipped with rail heads to conform to one of the basic configurations specified under 1.2 and the description as follows:

Type I - Single rail head with horizontal and vertical tool motions. The horizontal movement of the railhead shall be to the right of the table center and shall extend past the table center to the left.

Type II - Single rail head and side head, both with horizontal and vertical tool motions. The horizontal movement of the rail head shall be to the right of the table center and shall extend past the table center to the left.

Type III - Two rail heads, to provide vertical and horizontal tool motions for both. The horizontal feed movement of the rail heads shall be from the table center to the left and right, respectively.

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Type IV - Two rail heads and a side head with horizontal and vertical motion of the rail heads and side head. Rail head movements are the same as type III.

3.4.1 Base and column(s). The base and column(s) shall be an integrally cast structure of close grained cast iron or a steel weldment, heavily ribbed and braced to minimize deflection, distortion and vibration. The base shall contain the table drive mechanism and have means for securing and leveling the machine to a foundation. The column shall have precision finished vertical guideways for supporting and guiding the crossrail member.

3.4.2 Work holding table. Unless otherwise specified (see 6.2.1), the machine shall have a plain type work table with four faceplate type chuck jaws, having radial and parallel T-slots for mounting and positioning the chuck jaws at 90° positions, and for other type work holding devices. The T-slot dimensions shall conform to the applicable requirements of ANSI B5.1. The chuck jaws shall be designed for both internal and external grip on the work part. The table or chuck carrier shall be supported by anti-friction radial and thrust bearings or a single combination bearing incorporating the performance features of both types. Rotation shall be in both directions about the vertical axis. The table drive unit shall be of the infinitely variable speed type, the speed being adjustable in all ranges while under cutting conditions. Turning speed shall be automatically controlled to maintain a constant cutting rate (SFM) as diameter changes when turning or facing. The drive mechanism shall be protected from damage by chips, dirt, fluids, and from overload cutting conditions.

3.4.3 Crossrail. The crossrail shall be adjustable vertically on the column by power operated elevating screws. The elevating mechanism shall be designed so that parallelism to the table surface is maintained. When specified (see 6.2.1), the rail shall be automatically locked in position by power actuated clamping devices upon completion of a programmed move. This mechanism shall be electrically interlocked with the elevating drive to prevent the application of power to a clamped system. Safety devices shall be provided to prevent damage by overtravel at both upper and lower travel limits, and to prevent contact of the rail with the sidehead when so equipped. All main railways on which the railhead saddle bears shall have a hardness of not less than Rockwell C-58 for rolling contact bearing and not less than Rockwell C-50 for sliding contact bearing. When two railheads are provided, the rail shall provide the stability to permit precision finish cuts when operating both heads simultaneously.

3.4.4 Crossrail ram and turret heads. On each machine having two railhead units, the travel of each railhead unit shall be from maximum swing capacity on its side of the machine, to past the center of the table. When the type machine required is for a combination of ram and turret railhead

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units, the turret head shall operate on the right hand end of the crossrail. Unless otherwise specified, the ram head shall be of the nonindexing type for holding standard fixed tool holders. When specified (see 6.2.1), the ram shall be equipped with a power indexing turret for holding not less than four tools, and indexing about a vertical axis. The turret head unit shall be equipped with a power indexing turret for holding not less than five tools, and shall index about an essentially horizontal axis. Both type turrets shall be designed to accept holders for preset or qualified tooling. The indexing mechanism for each turret shall include a curvic coupling, or similar device providing equal positioning accuracy and positive locking features. Safety limit stops shall prevent overtravel of a unit, and collision of two units on the rail. Adjustable gibs, wear plates, or other means shall be provided to compensate for wear on sliding surfaces of rams and saddles. Suitable way wipers shall prevent entrance of dirt and chips between sliding units.

3.4.5 Side head. The side head unit, if supplied, shall be mounted on the right-hand side of the machine. The ram shall be equipped with a four position power indexing turret accepting preset or qualified tooling. The turret indexing mechanism shall have a curvic coupling, or similar device providing equal positioning accuracy and positive locking features. Adjustable gibs, wear plates or other means shall be provided to compensate for wear on sliding surfaces of the ram and saddle. The side head shall be capable of simultaneous motions on its two axes for positioning and contour turning. Safety limit stops shall prevent overtravel of moving units and interference between the side head and crossrail. Way wipers shall prevent entrance of dirt and chips between sliding units. The side head shall be counterbalanced.

3.4.6 Feed drive units. Each railhead and the side shall have an independent servo drive for each axis of movement. The drive units shall provide infinitely variable feed rates and rapid travel motions in both directions of travel on each driven axis. The feed components shall be driven by precision, preloaded recirculating ball nut and feed screw with zero backlash or provisions for adjustment to zero backlash or automatic backlash take-up. Feed screws shall be protected from damage by dirt and chips. The feed and positioning motions shall be accurately monitored by a closed loop feedback system for each axis. It shall be possible to adjust the feed rate while cutting to obtain optimum conditions. Automatic acceleration and deceleration shall provide safe start and stop speeds, and accuracy in positioning. Suitable protective means shall be provided to prevent damage to the drive mechanisms by cutting overload or obstruction to the driven mechanism.

3.4.7 Tool changer. When specified (see 6.2.1), the machine shall have an automatic tool changer with a magazine that is capable of storing not less than the tool capacity required for the style machine specified. The magazine shall be of the rotary type or any other closed loop recirculating configuration with the capability of presenting a preprogrammed tool to the tool changer for exchange with the tool held in the head. Tool change shall be accomplished by means of a power actuated mechanism capable of either interchanging a tool in the head with a tool selected from the tool magazine

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or removing the tool from the head and then inserting a tool from the magazine. The magazine shall permit convenient loading and unloading of tools, during operational setup. The tool changer shall be arranged for random tool selection which shall include the capability of reusing a tool that has been returned to the magazine after a program operation.

3.4.8 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.9 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.10 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79, and each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.10.1 Motors. Motors shall be rated for 60 minute duty cycle and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.4.11 Numerical control. The turning center shall have a fully automatic solid state soft-wired, integrated circuit type numerical control, complying with the requirements or EIA standard RS-281. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, linear axes movement, table operation, tool change, and other part program directed functions for turning centers of the type specified herein. Controlled axes shall be identified in accordance with EIA Standard RS-267. Control features shall include linear and circular interpolations, buffer storage, programmable interface, fixed cycles, part program storage, part program and buffer edit capability and control diagnostics to board level. The MCU shall direct machine functions from part program data stored in memory, manually input or from the media specified herein. The control shall initiate a halt or error signal should a fault condition occur in the control. All necessary executive program routines shall be furnished in the form of 1-inch 8-channel binary coded decimal punched tapeor 8-inch flexible

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disk for controls having volatile memory. The control unit shall conform to EIA Standards RS-281. A line voltage of ±10 percent from normal shall not adversely affect the numerical control system function. The control shall be capable of functioning in ambient temperatures ranging from 50° to 120° F and humidity ranging from 5 percent to 95 percent noncondensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. Control resolution shall be 0.0001 inch or 0.001 mm for each linear axis and not more than 0.001 of a degree for each rotary axis. When binary cutter location (BCL) data is specified (see 6.2.1), the numerical control shall have the necessary hardware and software to perform all geometry dependent functions which are ordinarily performed by the postprocessor.

3.4.11.1 Machine control unit (MCU). The MCU shall provide full control of each tool slide unit, tool turret indexing, table or chuck rotation, feed and speed selections, operating modes, tool changer (if provided) and other program directed functions for accomplishing boring and turning operations to the tolerances specified herein.

3.4.11.2 Data input.

3.4.11.2.1 Punched tape data input. Unless otherwise specified (see 6.2.1), the input media for the control shall be 1-inch, 8-channel punched tape. The data input format and codes shall comply with EIA Standards RS-274 and RS-358. The control shall be equipped with a photoelectric tape reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard feed and take-up spools 5-1/4 inches in diameter. The tape reader shall be housed in a dust free area of the control panel cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244
b. Even parity per EIA RS-358.
c. Misaligned sprocket holes
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.11.2.2 Flexible diskette data input. When 32-Bit Binary Cutter Location (BCL) Exchange Input for Numerically Controlled Machines is specified (see 6.2.1), the input media shall be an 8-inch flexible diskette recorded in accordance with 3.4.11.2.2.1. The input data format shall comply with EIA Standard RS-494. Functions 10.1, 10.2 and 10.3 of section 10, EIA-RS-494 shall apply. Manufacturers providing functions or features that cannot be addressed by command codes currently covered by EIA-RS-494 shall apply to

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the Electronic Industries Association, IE-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the new features, or functions. The control shall be equipped with an 8-inch flexible diskette reader. The reader shall have the capability to perform a cyclic redundancy check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.11.2.2.1 Flexible diskette format. When an 8-inch flexible disk is specified as the input media to the numerical control, it shall be a single-density recording on one side using a soft-sectored format. The disk shall be divided into 77 distinct locations on the single-side disk. Tracks shall be assigned numbers from 00 through 76. Track 00 shall be nearest to the edge of the disk and track 76 nearest the center. Tracks 00, 74, 75 and 76 shall not be used. There is an index hole in the jacket and the disk that shall be used for timing functions when a beam of light passes through it. Recording shall be accomplished by rotating the disk inside its jacket exposing the disk at the read/write head slot cut in the jacket. Data shall be recorded only on the side of the disk that is exposed to the read/write head. Each track shall be divided into 26 addressable sectors which contain 128 bits. All sectors shall be identified by the file starting sector number with track 00, sector 1, designated as 0. The first part program shall be located at sector 1E hexidecimal corresponding to track 1, sector 5. Where two byte numbers are referenced the high order byte shall be recorded first and the low order last. The data stored in a sector is called a record. Track 1, sectors 1 through 4 shall contain the program directory of up to 16-part program names in ASCII. Each file shall be packed on the disk in the same order as its directory entry. Unused directory entries shall point to the next available sector after the last file and contain the 30-character ASCII space characters. The last sector of each file shall be padded with end-of-file (EOF) records (8000 000). Diskette addressing shall comply with the requirements of table IV.

3.4.11.2.3 Manual data input (MDI). A means for manual data input (MDI) shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.11.3 Data display. A cathode ray tube (CRT) capable of displaying at least 250 characters simultaneously shall be provided. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, display the following data:

a. Four digit block sequence number.
b. Active block of stored data.
c. Actual position of each axis.
d. Feedrate and override condition.
e. Control and system errors messages.

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3.4.11.4 Control system modes of operation. The operational modes of the NC control shall include manual, automatic, single block, MDI, sequence number search, and optional stop.

3.4.11.5 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than plus or minus 99.9999 inches or 999.999 mm. The control shall be capable of automatically controlling the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall not be more than 0.0001 inch or 0.001 mm.

3.4.11.6 Absolute/incremental input. The control shall have absolute/incremental input capability. This mode shall be selected by a preparatory G-code which may be switched during the tape program. Not required if BCL input is specified (see 6.2.1).

3.4.11.7 Automatic tape code recognition. The control shall have the capability to automatically recognize and read tape punched with either EIA RS-244 or RS-358 (ASCII) character codes. Not required if BCL data input is specified (see 6.2.1).

3.4.11.8 Switchable inch/metric input. The control system shall provide switchable inch/metric input capability.

3.4.11.9 Part program storage. Unless otherwise specified (see 6.2.1), part program storage shall be in either solid-state or bubble-type memory. The MCU shall be capable of storing in memory the content of not less than 250 feet of postprocessed part program tape data unless otherwise specified (see 6.2.1). Not applicable if BCL data input is specified (see 6.2.1).

3.4.11.10 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory, to hold the memory provided at full load for a period of not less than 70 hours when the primary power source fails.

3.4.11.11 Part program edit. A program edit capability shall be provided to allow block insertion, deletion, or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.11.12 Feed and speed overrides. The feed and speed override features shall provide means for manual adjustment of the feedrate between 0 and at least 100 percent of the programmed feed in increments not greater than 10 percent of the programmed feedrate, and the table speed between 50 and at least 100 percent of the programmed speeds in increments not greater than 10 percent, to compensate for variable machine conditions.

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3.4.11.13 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.11.14 Preparatory functions (G-codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274. If the BCL data input is specified, codes in accordance with EIA-RS-494 shall be applied.

3.4.11.15 Miscellaneous functions (M-codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274. If the BCL data input is specified, codes in accordance with EIA-RS-494 shall be applied.

3.4.11.16 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed data.

3.4.11.17 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions. The internal storage shall delay or store multiple blocks of data until required by the controlling devices.

3.4.11.18 Block-by-block read. The control shall have a block-by-block read function to provide for reading information one block at a time.

3.4.11.19 Cutter compensation. The control shall have a tool length offset feature to permit adjustments to the programmed tool data by means not less than $\pm$99.9999 inches and $\pm$999.999 mm. Offsets shall be revisable by MDI through the keyboard. Unless otherwise specified (see 6.2.1), there shall be one pair of offsets for each tool position on the machine.

3.4.11.20 Dwell. A programmable dwell feature shall be provided having a dwell sequence based on table revolutions or in time in seconds. The range of dwell shall be not less than one complete table revolution at the lowest programmed rpm.

3.4.11.21 Threadcutting. Single point threading capability shall be provided, when specified (see 6.2.1), for production of all standard straight and taper threads, multiple lead threads, and scroll or face threads. Threads lead shall be at least 0.9999 inches and 99.9999-inch programmable thread length. Both internal and external threading shall be possible. When specified (see 6.2.1), segmented lead and variable lead threading capabilities shall be provided.

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3.4.11.22 Reversal error compensation. Reversal error compensation shall be provided for each axis of motion to compensate for drive gear backlash when slide reversal occurs. A means shall be provided to input new compensation values when required.

3.4.11.23 Axis calibration. Error compensation shall be provided to correct repeatable errors determined to exist due to leadscrew and feedback device inaccuracies. When the ballscrew is used in the feedback loop, a means shall be provided to input new error compensation values when required.

3.4.11.24 Additional features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

a. Four quadrant programming.
b. Incremental and continuous jog for each axis.
c. Leading zero suppression or decimal point programming.
d. Inches per minute (ipm)/millimeters per minute (mmpm) programming in all modes of operation.
e. Part program tape edit keylock.
f. Inches per revolution (ipr)/millimeters per revolution (mmpr) programming.

3.4.11.25 Operator control panel. Unless otherwise specified (see 6.2.1), the control unit shall have a panel which provides at least the following control functions: Emergency Stop; Cycle Start; Optional Stop; Mode Selection; Feed Hold; Feedrate Override; Table Speed Override; Jog: Jog Direction; and EOB Stop.

3.4.11.25.1 Pendant control. When specified (see 6.2.1), a pendant control shall be furnished as an operator control panel. The pendant control shall be connected to the machine to enable the operator to control machine functions while in a close observation position. Unless otherwise specified (see 6.2.1), the pendant control shall be capable of the same functions as stated for the operator control panel in paragraph 3.4.11.25.

3.4.11.26 Reference zero. Reference zero may be established by moving each axis to a reference limit switch or home position to establish synchronization and reset all registers to zero.

3.4.11.27 Set zero. The control unit shall have a "set zero/set reference" feature to permit the specified axis register to be reset to zero establishing present axis location as zero or other reference number.

3.4.11.28 Program tryout. The control shall have a program tape tryout feature to permit new part programs to be read through the control without machine motions.

3.4.11.29 Operational software. Operational software shall be resident in the control and shall contain all logic necessary to effectively operate

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the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the controls, machine tools, part program, and operator's actions.

3.4.11.30 Programmable interface. A software program for accommodating the software interface between the numerical control and machine shall be provided in one of the following forms:

a. One-inch, 8-channel punched tape complying with EIA-RS-244.
b. Nonvolatile memory.
c. Magnetic cartridge style tape.
d. Magnetic disk.

3.4.11.31 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished in either the control as non-volatile memory or in the form of 1-inch, 8-channel punched tape complying with EIA-RS-244 or 8-inch flexible disk as appropriate for the input media specified (see 6.2.1) to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise, and display failures, at least to board level.

3.4.11.32 Peripheral equipment interface. Unless otherwise specified (see 6.2.1), the control shall be equipped with an interface complying with EIA-RS-232.

3.4.11.33 Axis jog. The control panel or pendant shall have manual activated controls for single and continuous, low and high speed jog movements of the slides for each controlled axis in both the plus and minus directions. Incremental jog movements shall include steps of at least .0001, .001, .01, 0.1, and 1.0 inches.

3.4.11.34 Auxiliary live tools. When specified (see 6.2.1), the control shall be capable of controlling the rotation of a live tool as well as linear axes control and table rotation.

3.4.11.35 Auxiliary functions. Unless otherwise specified (see 6.2.1), the CNC shall respond to the following program commanded auxiliary functions: interpolation, fixed cycles, and dwell. Unless otherwise specified (see 6.2.1), miscellaneous functions shall include program stop, optional program stop, end-of-program, table CW, table CCW, table off, coolant on-off, tool change, and end-of-program (rewind). System preparatory and miscellaneous functions shall be coded in accordance with EIA RS-274.

3.4.11.36 Canned cycles. As specified (see 6.2.1), one or more of the following canned cycles shall be provided:

a. Taper generation.
b. Threadcutting.
c. Face cutting.
d. Drilling
e. Probing
f. Grooving

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3.4.11.37 Tape punch unit. When specified (see 6.2.1), the system shall have a tape punch unit for producing punched tapes from edited programs of tape being input and for programs stored in data memory. The unit shall have a plug receptacle and circuity that is compatible with the tape punch output of the system, punching both paper and mylar tapes of dimensions, characteristics and code in accordance with 3.4.11.2. The punching speed of the unit shall be not less than 60 characters per second.

3.4.11.38 Postprocessor. When specified (see 6.2.1), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified programming language (see 6.2.1) to the form and format required to operate the machine tool and control. The postprocessor shall be written for use on the type computer specified (see 6.2.1). Documentation sufficient to implement and utilize the postprocessor shall be provided for computer compatibility verification and validation at least 30 days prior to delivery of the machining center.

3.4.11.39 Interactive graphics software. When specified (see 6.2.1), the control shall be programmable in the MDI mode by use of a graphically symbolized keyboard for such steps as but not limited to angles, radii, single point contours, threads, bevels, chamfers and grooves. The CRT shall display the programmed moves as they are entered.

3.4.11.40 Conversational programming. When specified (see 6.2.1), the control shall be programmable by means of plain English commands through a prompting or menu driven format.

3.4.11.41 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.11.42 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary nominal voltage of 120 volts or less.

3.4.11.43 Probing. When specified (see 6.2.1), the machine shall be equipped with an NC controlled probe system for "in-process" gauging and tool probing. The part probe shall be loaded into the tool changer magazine and selected as required by the part program. The tool probe system shall be an integral part of the machine.

3.4.11.44 Adaptive control. When specified (see 6.2.1), the machine shall be equipped with adaptive control, enabling the optimization of feeds and speeds by monitoring actual cut horsepower versus maximum programmed cut horsepower and making adjustments accordingly. If the limits of SFM and IPR cannot be satisfied then the control shall initiate feedhold and generate an error message on the CRT.

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3.4.12 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with the machine.

a. A load meter installed on the machine or MCU console that constantly registers the horsepower load (horsepower units or percent of load) on the table rotational drive motor.

b. One set of special wrenches, operating, and repair tools that are normally furnished by the manufacturer.

3.4.13 Optional equipment. The following optional equipment shall be furnished as specified (see 6.2.1):

a. A worklight, conveniently mounted to illuminate the work area, complete with protective shield.

b. An elapsed time indicator of the non-reset type, with digital readout and designed to resist the effects of vibration, voltage fluctuation, temperature change, and humidity. The indicator shall be mounted on the machine in a position visible from the operator's station and be connected to record and accumulate machine on time to not less than 9999.9 hours.

c. Tools, tool holders, and adapters as specified.

d. Other optional equipment as specified.

3.5 Sizes and capacities. Unless otherwise specified (see 6.2.1), the sizes and capacities of the turning center shall meet the requirements in table I as applicable for the machine size specified in the contract or order.

3.6 Performance. The machine and numerical control system combination shall be capable of performing as specified herein. In addition, the machine shall be capable of performing drilling, threading, and boring to accomplish the capacities in table I while meeting the accuracies in table II.

3.7 Alignment. The machine and numerical control system shall meet the alignment and accuracy tolerances in table II.

3.8 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.9 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface and clearly legible. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the

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use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental test and cleaning procedures specified for the item.

3.9.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.9.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine and contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.11 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

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3.12 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspec-

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tion shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspections shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated in accordance with the machine tool builders' standard operating test procedure to check the machine's response to operating controls and data, as directed through the control unit. Tests shall be made in the operating mode to prove operation of control settings, power operated components , safety overtravel limit switches and stops, manual data input and readout systems, and miscellaneous functions as provided. Should a malfunction occur while operating, the test shall be re-run, after correction, until the machine can be fully cycled without a malfunction.

4.7.2 Alignment tolerance test. The machine shall be checked in accordance with the builders' standard test methods, using instruments having accuracies traceable to the Bureau of Standards. Tests shall be made to determine the machine alignments and slide positioning accuracies are within the tolerances of 3.6. Positioning accuracies shall be checked by using manual data input controls on the operator's control panel, for operating each tool slide. Positioning accuracy and repeatability shall be evaluated in accordance with NMTBA Definition and Evaluation of Accuracy and Repeatability for Numerically Controlled Machine Tools, Linear Definition. At least four randomly spaced target points, covering full travel of each slide, shall be used to evaluate accuracy.

4.7.3 Cutting test. A test part shall be machined to determine the machine's ability to maintain the cutting tolerances of 3.6 under actual machining conditions. The configuration of the test part, type of materials, and tooling shall be the supplier's choice to most effectively utilize all major capabilities of the machine being tested. The test part may be turned end-for-end (180°) to perform various types of cuts, or to check repeat accuracy of the same types of cuts. The test shall be designed to use each tool slide and to perform each type cut of which the machine is capable, such as straight, facing, taper, boring, concave and convex arcs, and threading cuts as applicable. Speed and feed rates shall be based on the recommended cutting speed (SFM) for the material and tools used, to obtain the best cutting conditions.

4.8 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9, in accordance with requirements of MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

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5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for use in any production shop where vertical turning, drilling, threading, or boring operations by MDI or CNC are performed.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size machine required, (see 1.2).

c. First article, if required, (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system, (see 3.2.1).

e. Specify measuring system graduation, US or SI (see 3.2.1.1).

f. Specify an additional safety and health requirements, (see 3.2.5).

g. Painting requirements, if different, (see 3.3.5).

h. Specify if electromagnetic interference control is required (see 3.3.9).

i. Work holding table requirements, if different (see 3.4.2).

j. Rail clamping device(s), if required (see 3.4.3).

k. Specify requirements for ram turret, if applicable (see 3.4.4).

l. Requirements for the tool changer, if desired (see 3.4.7).

m. Specify electrical system requirements, if different (see 3.4.10).

n. Motor enclosure, if different (see 3.4.10.1).

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o. Specify EIA-RS-494 binary cutter location data input, if required (see 3.4.11).

p. Specify input media, if different (see 3.4.11.2.1).

q. Tape reader, take-up spool sizes, if different (see 3.4.11.2.1).

r. Specify data display requirements, if different (see 3.4.11.3).

s. Specify part program storage media, if different (see 3.4.11.9).

t. MCU memory capacity, if different (see 3.4.11.9).

u. Requirements for memory backup power, if different (see 3.4.11.10).

v. Tool offset data requirements, if different (see 3.4.11.19).

w. Single point threadcutting capability, if required (see 3.4.11.21).

x. Segmented lead threadcutting, if required (see 3.4.11.21).

y. Additional numerical control features, if required (see 3.4.11.24).

z. Specify pendant control and functions, if required (see 3.4.11.25.1).

aa. Specify MCU interface characteristics, (see 3.4.11.30).

bb. Specify diagnostics system software format, (see 3.4.11.31).

cc. Peripheral equipment interface, if different (see 3.4.11.32).

dd. Specify requirements for controlling auxiliary live tooling, if required, (see 3.4.11.34).

ee. Specify auxiliary and miscellaneous functions (see 3.4.11.35).

ff. Canned cycles, if different (see 3.4.11.36).

gg. Tape punch unit, if required, (see 3.4.11.37).

hh. Postprocessor requirement, if any, (see 3.4.11.38).

ii. Graphics software, if required (see 3.4.11.39).

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jj. Specify form and format for conversational programming capabilities, if required (3.4.11.40).

kk. Probing, (see 3.4.11.43).

ll. Adaptive control (see 3.4.11.44).

mm. Standard equipment to be supplied, if different (see 3.4.12).

nn. Optional equipment as required (see 3.4.13).

oo. Sizes and capacities other than from table I (see 3.5).

pp. Requirements for repair parts, if any (3.8).

qq. Whether or not a lubrication chart is required (see 3.9.1).

rr. Whether or not a nameplate is required (see 3.9.2).

ss. Whether or not an hour meter is required (see 3.10).

tt. Requirements of a first article examination (see 4.4).

uu. Requirements of the quality conformance inspection (see 4.5).

vv. Specify preservation, packing and marking requirements (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machine in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirement that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation

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guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, since the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(f).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (key word listing.

Binary Cutter Location (BCL)
Computer Numerical Control
Flexible Diskette
Punched Tape
Vertical Turning Center

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TABLE I. Sizes and capacities.

Characteristic	(Requirements in inches) 1/ Size 36	Size 48	Size 60	Size 72	Size 84	Size 96
Table, chuck diameter	36 - 46	47 - 56	57 - 66	67 - 77	78 - 95	96 - 118
Swing above side head	48	56	67	77	96	108
Swing, side head in work position	37	49	59	69	90	102
Maximum turning height	40	48	55	62	70	88
Ram slide vertical travel	38	39	47	47	47	60
Turret slide vertical travel	30	30	30	30	30	30
Turret head or ram travel right of center, single head	26	32	38	48	54	60
Travel past center, single head	4	4	4	4	4	4
Side head vertical travel above table top	30	38	46	52	52	66
Side head ram travel, horizontally	18	24	24	24	24	30
Table speed ram RPM	5 to 375	4 to 250	4 to 235	3 to 200	3 to 200	2 to 130
Feed range, manufacturer's standard	<---------- .1 to 250 I.P.M. ---------->					
Traverse rate, maximum	<---------- 250 I.P.M. or .0005 to 2.0 I.P.R. ---------->					
Main drive motor H.P.	<---------- 50 ---------->					
Optional H.P.	<---------- 75 or 100 ---------->					

1/ Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size as shown in above table. When the largest size and capacity machine shown in the above table is ordered, the size and capacity of the machined offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

** One (1) Hour duty ratings.

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TABLE II. Accuracy tolerances.

Accuracy check	Tolerance in inches	Optional tolerances, when specified under 3.6
Alignment accuracies:		
Table flatness, concave only	0.0005 per 12 inches	—
	0.002 maximum overall	—
Table radial runout	0.0007 TIR	—
Table axial runout	0.0007 TIR	—
Table center bore runout	0.0007 TIR	—
Crossrail parallel to plane of table rotation	0.0006 per 12 inches	—
Rail head slide or ram vertical travel		
Parallel to table axis, right to left	0.0005 per 12 inches	0.00025 per 12 inches
Total tolerance	0.00025 more each additional 12"	0.0005 more per 36 inches
Front to back, total	0.0015 per 12 inches	0.001 per 12 inches
Side head vertical travel parallel to table axis	0.0005 per 12 inches	0.00025 per 12 inches
Side head ram travel parallel to table plane of rotation	0.0005 per 12 inches	0.00025 per 12 inches
Total tolerance	0.0002 more each additional 12"	—
Turret indexing, repeatability	±.3 second arc	—
Positioning accuracies:		
Linear positioning each axis, uni-directional	0.0005 in 12 inches	±0.001 per 36 inches
Linear positioning repeatability	0.0005	—
Cutting accuracies:		
Deviation from tool path, linear and circular	±0.002	—
Roundness, checked at any point or internal or external diameter	+0.001	—
Cylindrical surface taper not more than	0.001 per 12 inches	—
Squareness, face of shoulder to turned or bored surface	0.001 per 12 inches	—
Angle of taper, deviation from program not to exceed	±1 minute of arc	—
Overshoot or undercut	±0.001 inch	—
Surface finish not rougher than	60 microinches	—

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AR
Navy - YD
Air Force - 84
DLA - GS

User activities:
Army - As

Preparing activity:
DLA - IP

Project (3411-0064)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-80248A	Boring and Turning Machines, Vertical, Computer Numerical Control

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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